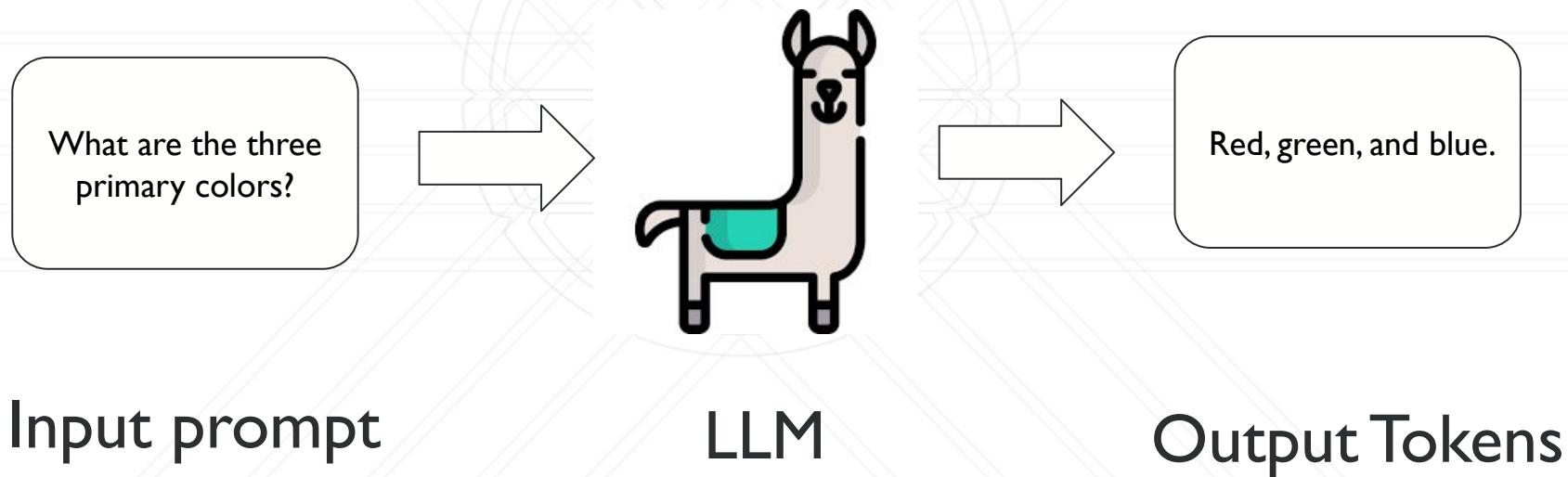


Distributed Deep Learning on GPU-based Clusters

Prajwal Singhanian, Lannie Dalton Hough, Cunyang Wei
Department of Computer Science

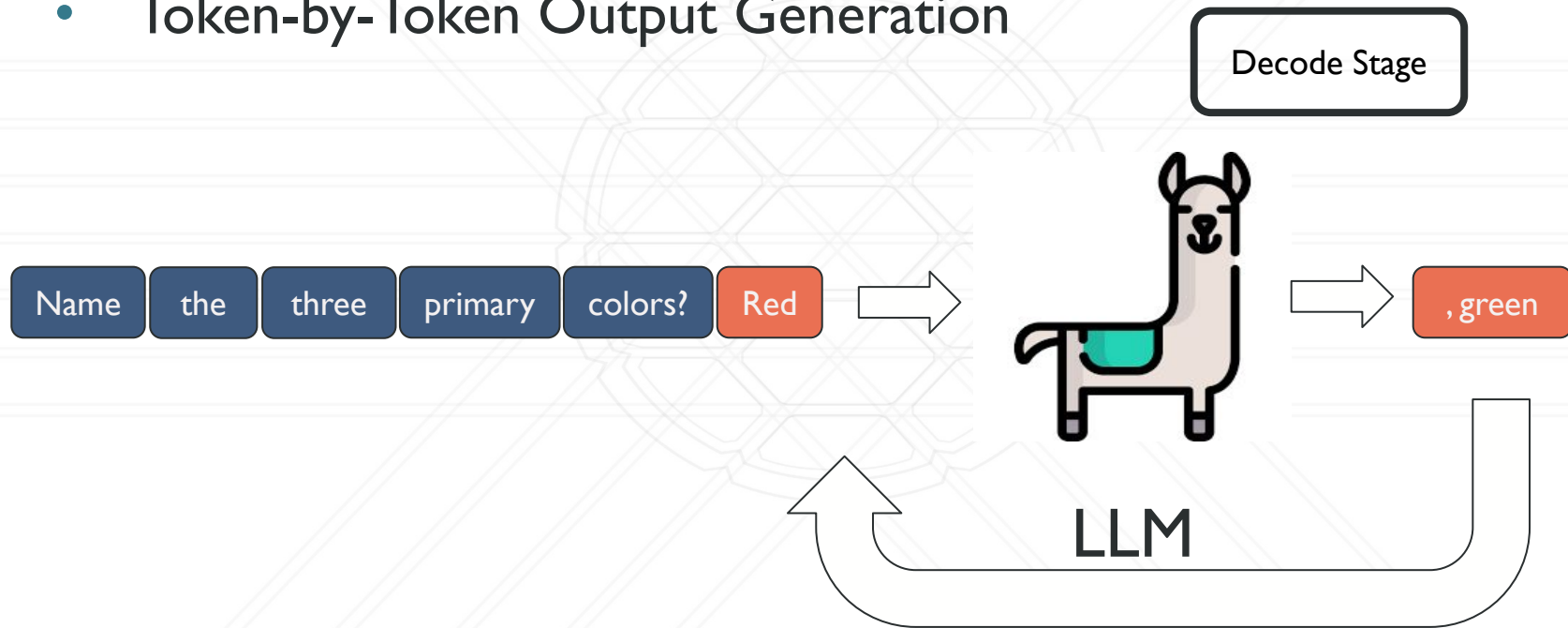
Inference

- Generating outputs from a trained language model



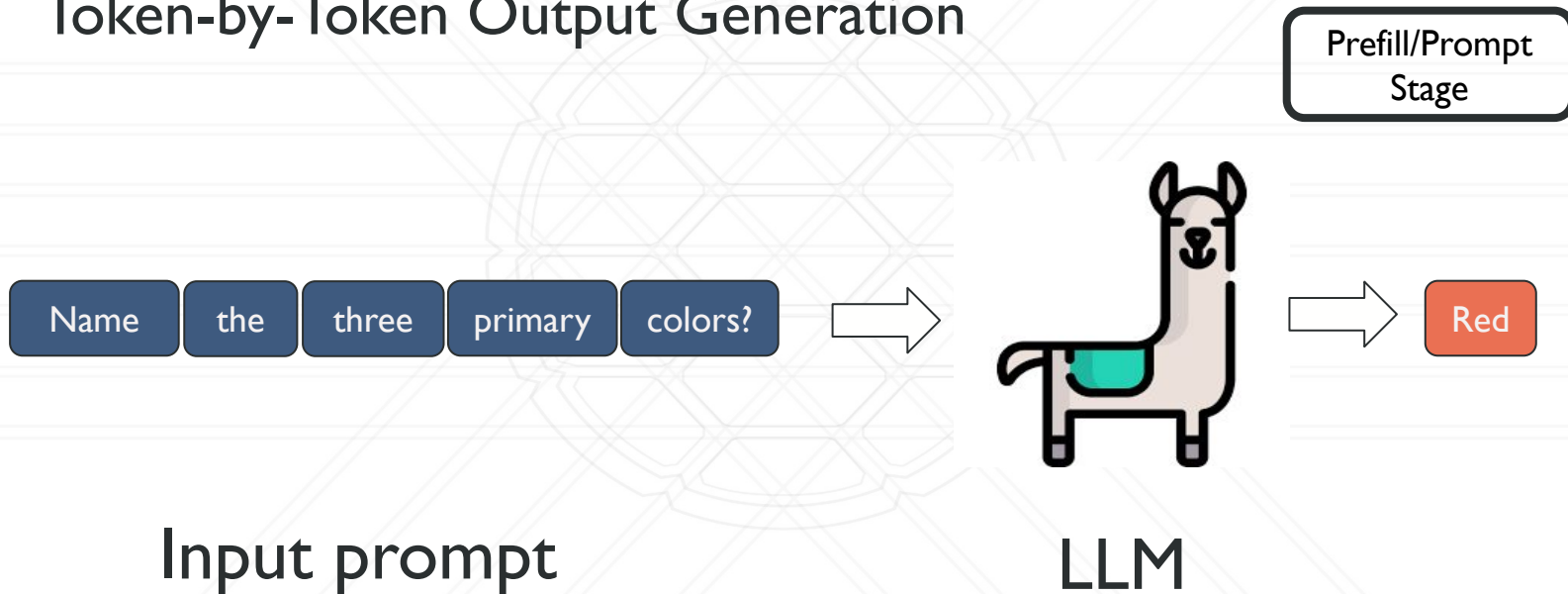
Autoregressive LLM Inference

- Token-by-Token Output Generation



Autoregressive LLM Inference

- Token-by-Token Output Generation



Types of Inference

- Online Inference

- Generating outputs in real time as user input is received.

- ChatGPT, Claude

- Offline inference

- Generating outputs on pre-collected data

- Synthetic data generation, benchmarking

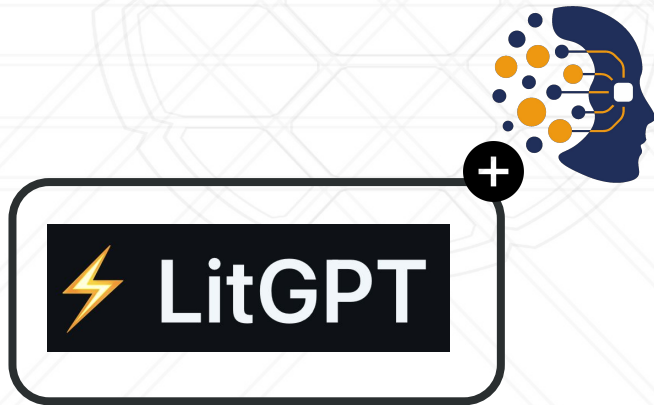
Inference Frameworks

- Why is inference different from training to require separate optimizations/frameworks?
 - **Auto-regressive** nature of inference
 - Memory Bound!
 - No backward pass in inference
 - Different workload characteristics, eg. smaller batch size

Inference Frameworks

- What does an inference framework provide you?
 - Model Implementations with/without optimized kernels
 - Memory and request management

 VLLM



Hugging Face

Hands-On: Offline Inference

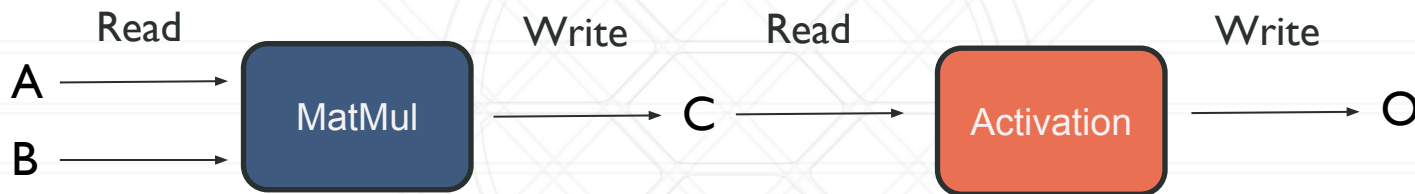
- Code – `infer.py`

```
CONFIG_FILE=configs/inference_yalis.json sbatch infer_single.sh
```

- Try adding your own prompts to
`data/inference/prompts.txt`

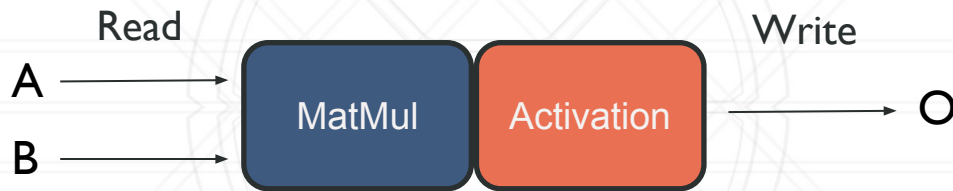
Optimized Kernels with Torch Compile

- Kernel Fusion



Optimized Kernels with Torch Compile

- Kernel Fusion



Hands-On: Torch Compile

- Open `infer_single.sh` and set:
`YALIS_DISABLE_COMPILE=0`

```
CONFIG_FILE=configs/inference_yalis.json sbatch infer_single.sh
```

CudaGraphs to Reduce Launch Overheads

- CUDA Graphs: Create a computation graph to launch multiple kernels at once

Without Cuda Graphs



With Cuda Graphs



Hands-On: Cuda Graphs

- Open `infer_single.sh` and set:
`YALIS_DISABLE_DECODE_CUDAGRAPH=0`

```
CONFIG_FILE=configs/inference_yalis.json sbatch infer_single.sh
```

Hands-On: Distributed Inference

```
CONFIG_FILE=configs/inference_yalis.json sbatch infer_multi.sh
```

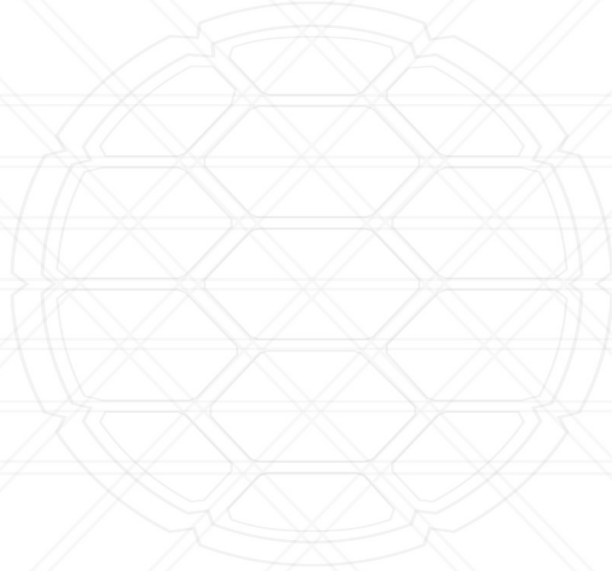
Online Inference using VLLM

- State-of-the-art framework for inference
 - Inference-optimized GPU kernels for LLMs
 - Paged Attention
 - Tensor Parallelism
 - Optimized for high-throughput scenarios

Hands-On: Online Inference

- We have setup a server on one of the compute nodes

```
./llm_request.sh <VLLM IP> "San Francisco is a" 64
```

Prajwal Singhanian, Lannie Dalton Hough, Cunyang Wei

Department of Computer Science

prajwal@umd.edu, cunyang@umd.edu, ldhough@umd.edu

